

PHASE THEORY — MATHEMATICAL FORMALIZATION FRAMEWORK

*Companion Paper to “Phase Theory: A Unified Theory of Matter, Light, and
Geometry”*

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Abstract

This paper provides the complete mathematical formalization framework required to elevate Phase Theory from a physical ontology into a scientifically admissible physical theory. We present: (1) a rigorous axiomatic foundation consisting of three foundational axioms—Phase Primacy, No Collapse, and the Admissibility Constraint—derived from the five axioms of the parent theory [1]; (2) the complete field-theoretic formulation, including the action functional, Euler-Lagrange equations, and gauge structure on a principal fiber bundle; (3) formal recovery of quantum mechanics via emergent wavefunction construction and derivation of the Schrödinger equation from the phase-relaxation equation in the slowly varying background regime; (4) Bell compatibility without superluminal signaling, achieved through global admissibility enforcement on joint phase configurations; (5) measurement resolution without collapse postulates, grounded in topological decoherence and coarse-graining over admissible phase classes; (6) scientific admissibility criteria including renormalizability or UV completeness, emergent Lorentz invariance, Standard Model recovery, and General Relativity recovery via the Jacobson thermodynamic argument; and (7) seven classes of falsifiable predictions distinguishing Phase Theory from standard QM/QFT/GR. Phase Theory becomes a scientific theory—as distinct from a coherent ontological reinterpretation—only upon experimental confirmation of at least one predicted deviation from established physics.

Keywords: phase ontology, topology, emergence, gravity, quantum foundations, mathematical formalization, admissibility constraint, topological defects, Bell inequality, decoherence

0. Scientific Admissibility Framework

We begin with an epistemological preamble. A physical theory cannot be “proven” true in the mathematical sense. It can only be:

1. Formally axiomatized;
2. Shown mathematically consistent;
3. Reduced to known physics in appropriate limits;
4. Made experimentally falsifiable;
5. Empirically tested.

This document outlines the necessary formal structure required to elevate Phase Theory into a scientifically valid physical theory. The framework follows the tradition of Popper’s falsificationism [2]: a theory is scientific if and only if it specifies conditions under which it would be refuted. We further adopt the methodology of Lakatos [3], treating Phase Theory as a scientific research programme possessing a hard core (the three axioms of Section 1), a protective belt (the specific action functional, coupling constants, and approximation schemes of Sections 2–6), and a heuristic (the seven prediction classes of Section 7) that generates novel empirical content.

Phase Theory, as formulated in the parent preprint Hanks (2026) [1], satisfies criteria (1)–(4) and specifies experimental protocols for (5). The present paper makes this claim precise by constructing the required mathematical apparatus and enumerating the falsifiable predictions that distinguish the theory from all extant alternatives.

1. Foundational Ontology

The foundational ontology of Phase Theory is specified by three axioms, derived from the five axioms of the parent theory [1]. These three axioms are logically independent: no subset implies any other (see Section 8.1 for formal verification). Together they define the ontic content of the theory.

1.1 Axiom 1 — Phase Primacy

Definition 1.1 (Phase Substrate). *The fundamental field is a continuous map*

$$\Phi : M \rightarrow T(1)$$

where M is a connected topological space (the *base space*), and T is a compact Lie group serving as the *target manifold*. For the Standard Model case, $T = \text{U}(1) \times \text{SU}(2) \times \text{SU}(3)$.

The base space M carries no metric, no causal structure, and no preferred dimensionality. It is equipped only with the structure of open sets and continuity—that is, a topology. It is emphatically *not* spacetime. Spacetime, together with its Lorentzian metric and causal structure, emerges from the dynamics of Φ (see Section 6). The target manifold T encodes internal degrees of freedom. The choice $T = \text{U}(1) \times \text{SU}(2) \times \text{SU}(3)$ yields the Standard Model gauge group as a structural consequence of the phase substrate, not as an externally imposed symmetry.

Axiom 1 (Phase Primacy). *All physical observables are functionals of Φ . The phase field Φ is the sole ontic primitive. Space, time, matter, light, and geometry are emergent structures derivable from the configuration and dynamics of Φ .*

The topological structure of the target manifold determines the classification of stable excitations via homotopy theory:

- $\pi_1(T) = \mathbb{Z}$ — vortex lines, identified with charged particles;
- $\pi_2(T) = 0$ — no stable monopoles in the Standard Model case;
- $\pi_3(T) = \mathbb{Z} \oplus \mathbb{Z}$ — instantons and skyrmions, identified with baryonic matter.

The principal fiber bundle $P(M, T)$ carries a connection one-form $A \in \Omega^1(P, \mathfrak{t})$ where $\mathfrak{t} = \text{Lie}(T)$ is the Lie algebra of the structure group. The curvature of this connection is

$$F = dA + A \wedge A \quad (2)$$

and the covariant derivative of the phase field is

$$D\Phi = d\Phi + A \cdot \Phi \quad (3)$$

The connection A and curvature F will be identified with the gauge fields and field strengths of the Standard Model in the appropriate limit (Section 6.3).

1.2 Axiom 2 — No Collapse Postulate

Axiom 2 (Global Unitarity). *All dynamics are globally unitary. The evolution of the phase substrate is governed by:*

$$\Phi(u) = U(u) \Phi(0) \quad (4)$$

where $U(u) = \exp(-iHu)$, H is the phase-consistency Hamiltonian derived from the variational principle (Section 2), and u is the *update parameter*. The update parameter is not physical time. Physical time emerges from the partial ordering of updates as established in Axiom 2 of the parent theory [1].

There exists no stochastic projection operator. No collapse postulate is admitted at any level of the theory. The appearance of probabilistic outcomes arises exclusively from coarse-graining over admissible phase classes, as formalized in Section 5.

Theorem 1.1. *The evolution operator $U(u)$ preserves the admissibility constraint $A[\Phi] = 0$ (Axiom 3). That is, if $A[\Phi(0)] = 0$, then $A[U(u)\Phi(0)] = 0$ for all u .*

Proof sketch. The Hamiltonian H is constructed from the phase-inconsistency functional $\mathbb{I}[\Phi]$, which is gauge-invariant and preserves topological sector. Since admissibility is defined by topological consistency (Definition 1.3 below), and unitary evolution preserves topological invariants—homotopy classes are invariant under continuous deformations generated by a continuous Hamiltonian flow—the admissibility constraint is preserved. $\square$

1.3 Axiom 3 — Admissibility Constraint

Axiom 3 (Admissibility). *Not all mathematically possible phase assignments are physically realizable. The admissibility functional*

$$A[\Phi] : \Gamma \rightarrow \mathbb{R}_{\geq 0} \quad (5)$$

where Γ is the configuration space of Φ , selects physically realized states by the condition:

$$A[\Phi] = 0 \quad (6)$$

The admissibility functional encodes global phase consistency. It replaces the collapse postulate of standard quantum mechanics with a global consistency constraint. The admissibility functional is related to the phase-inconsistency functional $I[\Phi]$ of the parent theory [1]:

$$A[\Phi] = 0 \Leftrightarrow \Phi \text{ minimizes } I \text{ within its topological sector}$$

Formally:

$$A[\Phi] = I[\Phi] - \inf_{\Phi' \in \sigma(\Phi)} I[\Phi'] \quad (7)$$

where $\sigma(\Phi)$ denotes the topological sector of Φ —the connected component of Γ containing Φ under continuous deformations that preserve homotopy class.

Definition 1.2 (Properties of A). The admissibility functional satisfies:

- (i) *Non-negativity:* $A[\Phi] \geq 0$ for all $\Phi \in \Gamma$.
- (ii) *Topological invariance:* $A[\Phi]$ depends only on the deviation from the sector minimum, not on continuous deformations within the sector.
- (iii) *Gauge invariance:* $A[g\Phi] = A[\Phi]$ for all gauge transformations $g \in G$.
- (iv) *Lower semicontinuity:* A is lower semicontinuous in the natural topology on Γ , ensuring that minimizers exist in each topological sector by the direct method of the calculus of variations.

This replaces the collapse postulate with global consistency. The constraint $A[\Phi] = 0$ selects the physically realized configurations from the space of all mathematically

possible ones. Collapse is not a physical process; it is the recognition that only admissible configurations are ontologically real.

2. Field-Theoretic Formulation

2.1 The Action Functional

The dynamics of the phase substrate are derived from the variational principle $\delta S = 0$ applied to the action functional:

$$S[\Phi] = \int_M L(\Phi, D\Phi, F) \, d\mu \quad (8)$$

where the Lagrangian density is given by:

$$L = \frac{1}{2} K_{ab}(\Phi) D_\mu \Phi^a D^\mu \Phi^b - V(\Phi) - \frac{1}{4\lambda} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) \quad (9)$$

Here $K_{ab}(\Phi)$ is the positive-definite *phase-stiffness tensor* (the metric on T pulled back to field space), $V(\Phi)$ is the *phase potential*, λ is the gauge coupling constant, $F_{\mu\nu}$ is the connection curvature, and $d\mu = \sqrt{(\det g)} \, d^4x$ where g is the emergent metric.

The three terms admit direct physical interpretation:

- **Phase-gradient energy** (kinetic term): $\frac{1}{2} K_{ab} D_\mu \Phi^a D^\mu \Phi^b$ penalizes rapid spatial and temporal variation of the phase field. This term drives the substrate toward smooth configurations.
- **Phase-potential energy**: $V(\Phi)$ encodes the energetic landscape of the internal target space. Minima of V correspond to vacuum configurations; excitations about these minima correspond to particles.
- **Connection-curvature energy**: $\frac{1}{4\lambda} \text{Tr}(F_{\mu\nu} F^{\mu\nu})$ is the Yang-Mills action for the gauge connection, governing the dynamics of the gauge fields that mediate interactions between phase excitations.

2.2 Euler-Lagrange Equations

The field equations follow from the variational principle $\delta S = 0$ with respect to independent variations of Φ and A .

Phase-relaxation equation (variation with respect to Φ):

$$D_\mu(K_{ab} D^\mu \Phi^b) - \frac{1}{2} (\partial K_{bd} / \partial \Phi^a) D_\mu \Phi^b D^\mu \Phi^c - \partial V / \partial \Phi^a = 0 \quad (10)$$

Gauge-field equation (variation with respect to A):

$$D_\nu F^a_{\nu\mu} + J^a_\mu = 0 \quad (11)$$

Equation (10) is the *phase-relaxation equation*, governing the dynamics of the substrate field Φ . It is a nonlinear wave equation on the target manifold T , generalizing the harmonic map equation to include gauge coupling and a potential. Equation (11) is the *gauge-field equation*, identified with the Yang-Mills equation sourced by the phase current J^a_μ . The current is constructed from the variation of the kinetic term with respect to A :

$$J^a_\mu = K_{ab} T^a_{cd} \Phi^c D_\mu \Phi^d \quad (12)$$

where T^a_{cd} are the generators of the gauge group in the representation carried by Φ .

2.3 Reduction to Klein-Gordon

Consider the special case of a single real scalar component with quadratic potential $V(\Phi) = \frac{1}{2} m^2 \Phi^2$, flat stiffness $K_{ab} = \delta_{ab}$, and a Minkowski background. The phase-relaxation equation (10) reduces to:

$$\square \Phi + m^2 \Phi = 0 \quad (13)$$

where $\square = \partial_\mu \partial^\mu$ is the d'Alembertian. This is the Klein-Gordon equation. Standard scalar quantum field theory thus appears as a limiting case of Phase Theory in the linearized, flat-background regime.

2.4 Gauge Redundancy

Two configurations Φ, Φ' are physically equivalent if they are related by a gauge transformation $g: M \rightarrow T$:

$$\Phi' = g \cdot \Phi, \quad A' = g A g^{-1} + g dg^{-1} \quad (14)$$

The physical configuration space is the quotient Γ/G , where G is the infinite-dimensional group of gauge transformations. All physical observables must be gauge-invariant functionals of Φ and A . The admissibility functional $A[\Phi]$ is gauge-invariant by construction (Property (iii), Section 1.3).

2.5 Topological Invariants

Topological sectors are classified by invariants that are preserved under continuous deformation and gauge transformation. The principal invariants are:

- **Winding numbers:** $\deg(\Phi) \in \pi_n(T)$. For the $U(1)$ factor, the winding number is $k = (1/2\pi) \oint d\Phi \in \mathbb{Z}$, identified with electric charge.
- **Linking numbers:** For extended defects (vortex lines, domain walls), the linking number $\ell(C_1, C_2) \in \mathbb{Z}$ classifies the mutual entanglement of defect cores.
- **Knot invariants:** For knotted defect cores, the Jones polynomial $V_K(t)$ and the HOMFLY polynomial $P_K(a, z)$ provide finer topological classification.
- **Holonomy classes:** $\text{Hol}(A, \gamma) = P \exp(\oint_\gamma A) \in T$, where P denotes path ordering. Wilson loops $W(A, \gamma)$ are gauge-invariant observables [16].

The complete topological label of a phase configuration is the tuple $(k, Q_H, \mathbf{t}, \chi, R)$, where k is the winding number, Q_H is the Hopf charge, $\mathbf{t}$ is the set of linking/knotting data, χ is the Euler characteristic of the defect manifold, and R is the representation label under T .

3. Recovery of Quantum Mechanics

3.1 Emergent Wavefunction

We define the emergent wavefunction for a localized phase defect propagating through a slowly varying background:

$$\psi(\mathbf{x}, t) = |\psi(\mathbf{x}, t)| e^{i\Phi(\mathbf{x}, t)} \quad (15)$$

where $\Phi(\mathbf{x}, t)$ is the local phase of the defect relative to the background, and $|\psi|$ is the amplitude determined by the defect profile. The wavefunction ψ is not a fundamental object; it is a derived quantity encoding the local phase and amplitude structure of a topological excitation of Φ .

3.2 Derivation of the Schrödinger Equation

Theorem 3.1 (Schrödinger Emergence). *In the regime of a well-localized defect propagating through a slowly varying phase-stiffness background, the phase-relaxation equation (10) reduces to:*

$$i\hbar \partial\psi/\partial t = -(\hbar^2/2m) \nabla^2\psi + V(\mathbf{x})\psi \quad (16)$$

where m is the phase-inconsistency energy of the defect (identified with inertial mass), $V(\mathbf{x})$ is the external phase-stiffness potential, and $\hbar$ is the characteristic scale of the stochastic drive inherited from the microscopic fluctuations of the phase substrate.

Proof sketch. Parameterize the defect by the polar decomposition $\psi = |\psi|e^{i\theta}$. Substitute into the phase-relaxation equation (10) in the regime where the background phase stiffness K_{ab} varies on scales much larger than the defect size. In this regime, the nonlinear terms involving $\partial K_{bd}/\partial\Phi^a$ become subdominant. Separating the resulting equation into real and imaginary parts yields, respectively, the quantum Hamilton-Jacobi equation and the continuity equation:

$$\partial\rho/\partial t + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (\text{continuity})$$

$$\partial S/\partial t + |\nabla S|^2/(2m) + V - (\hbar^2/2m) \nabla^2|\psi|/|\psi| = 0 \quad (\text{Hamilton-Jacobi})$$

where $\rho = |\psi|^2$, $\mathbf{v} = \nabla S/m$, and $S = \hbar\theta$. Combining these via the standard Madelung transformation [6] yields the Schrödinger equation (16). $\square$

3.3 Hilbert Space Structure

Proposition 3.1. *The space of small perturbations $\delta\Phi$ about a stable background Φ_0 forms a Hilbert space H with inner product:*

$$\langle \delta\Phi_1, \delta\Phi_2 \rangle = \delta^2 I[\delta\Phi_1, \delta\Phi_2] \quad (17)$$

where $\delta^2 I$ is the second variation of the phase-inconsistency functional evaluated at Φ_0 . This inner product is positive-definite (by the stability of Φ_0 , which requires that $\delta^2 I > 0$ for all nonzero perturbations) and the resulting space is complete (by the lower semicontinuity of I and standard Sobolev embedding theorems). The quantum-mechanical Hilbert space is therefore an emergent structure, not a fundamental postulate.

3.4 Born Rule from Phase-Space Measure Invariance

The Born rule emerges from phase averaging over topological sectors. Consider a system in a superposition of admissible sectors labeled by quantum numbers n :

$$\Phi = \sum_n c_n \Phi_n \quad (18)$$

The probability of finding the system in sector n upon measurement is:

$$P(n) = |\langle n | \psi \rangle|^2 = |c_n|^2 \quad (19)$$

The quadratic dependence follows from the quadratic form of the phase-inconsistency functional in Eq. (7): the natural measure on the space of admissible configurations is the Gaussian measure induced by I , which assigns weight proportional to $|c_n|^2$ to each sector. During measurement (decoherence), coherence between sectors is destroyed by environmental interaction (Section 5), and the relative probability of each outcome is proportional to the squared phase amplitude. This derivation parallels the Gleason and Zurek approaches [12] but is grounded in the specific structure of the admissibility functional.

4. Bell Compatibility Without Superluminal Signaling

4.1 Joint Phase Configuration

For a bipartite system consisting of subsystems A and B , the joint phase configuration is:

$$\Phi_{AB}: M_A \times M_B \rightarrow T(20)$$

where M_A and M_B are the base-space regions supporting subsystems A and B , respectively. The joint configuration Φ_{AB} is not, in general, a product $\Phi_A \otimes \Phi_B$. The deviation from product structure encodes entanglement.

4.2 Global Admissibility Enforcement

Theorem 4.1 (Bell Compatibility). *If global admissibility holds— $A[\Phi_{AB}] = 0$ —then: (i) correlations between A and B can violate Bell inequalities [11]; (ii) no superluminal signaling is possible.*

Proof structure.

Part (i): Admissibility enforces joint consistency through shared phase topology. The topological constraints linking the defect configurations of A and B —linking numbers, braiding phases, and correlated winding numbers—are non-separable: they cannot be decomposed into independent constraints on A and B individually. These non-separable topological constraints produce correlations that are non-classical and can violate the CHSH bound $|S| \leq 2$ up to the Tsirelson bound $|S| \leq 2\sqrt{2}$, exactly as in standard quantum mechanics.

Part (ii): Local operations on subsystem A consist of gauge transformations and local phase deformations. These preserve $A[\Phi_{AB}] = 0$ (by gauge invariance and topological sector preservation). Crucially, no local operator on A modifies the reduced marginal of B :

$$\text{Tr}_A(\rho_{AB}) = \rho_B \quad \forall \text{ local operations on } A \quad (21)$$

This is because the admissibility constraint is topological: it constrains the global linking structure but does not transmit information through the base space faster than the maximum phase propagation speed (the emergent speed of light c). Therefore: Bell violation is achieved without superluminal causation. $\square$

4.3 Entanglement as Shared Phase Topology

In Phase Theory, two subsystems are entangled if and only if they share non-separable topological connectivity—links, braids, or correlations between defect configurations that cannot be removed by local operations alone. Entanglement is not a mysterious “spooky action at a distance”; it is the residual topological structure of a shared phase history.

Proposition 4.1 (Monogamy of Entanglement). *Maximal entanglement between A and B precludes entanglement of A with a third subsystem C .*

This follows because the topological linking capacity of a defect is finite, bounded by the complexity of its invariants and the coherence limitation (Axiom 4 of the parent theory [1]). If the full linking capacity of A is saturated by its connection to B , no residual capacity remains for topological linkage with C . This reproduces the monogamy inequalities of standard quantum information theory.

5. Measurement Without Collapse

5.1 Measurement as Boundary Interaction

Measurement is defined as a coupling between a small phase subsystem S (the system being measured) and a large environment E (the apparatus plus its environment):

$$\Phi_S \otimes \Phi_E \rightarrow \Phi'_{SE} \quad (22)$$

subject to the global admissibility constraint $A[\Phi'_{SE}] = 0$. The interaction correlates the topological sector of S with a macroscopically distinguishable configuration of E . No special “measurement postulate” is required; the interaction is governed by the same phase-relaxation equation (10) that governs all dynamics.

5.2 Decoherence Mechanism

Theorem 5.1 (Decoherence). *When subsystem S interacts with environment E , phase coherence between different topological sectors of S is transferred to E and effectively lost. The mutual information between S and its distinct sectors saturates the coherence bound:*

$$\mathcal{J}(\Phi_S; \Phi_E) \leq C \cdot \min(|\partial S|, |\partial E|) \quad (23)$$

where $|\partial S|$ and $|\partial E|$ denote the boundary areas of the respective subsystems, and C is a constant related to the coherence scale. This is the area-scaling holographic bound inherited from Axiom 4 of the parent theory [1], consistent with the Bekenstein–t Hooft–Susskind entropy bounds [8, 9, 10].

5.3 Apparent Collapse

Apparent probabilistic outcomes arise from coarse-graining over admissible phase classes. When a measurement interaction transfers the inter-sector coherence of S into the vast phase space of E , the reduced state of S becomes effectively diagonal in the basis of topological sectors. The observer, having access only to S and a macroscopic record in E , perceives a definite outcome selected with probability given by the Born rule (Section 3.4). No stochastic projection operator is needed; no physical discontinuity occurs.

5.4 Preferred Basis Selection

The preferred basis is selected by topological structure. Decoherence is most rapid for sectors that are maximally separated in configuration space, which are precisely those labeled by topological invariants $(k, Q_H, \mathbf{t}, \chi, R)$. Sectors with distinct topological labels decohere on timescales much shorter than those with the same topological label but different smooth configurations. This mechanism is consistent

with Zurek’s einselection programme [12] but is grounded in the topology of the phase substrate rather than postulated independently.

6. Scientific Requirements for Validity

For Phase Theory to qualify as a scientific theory, it must satisfy the criteria enumerated in Section 0. We address each in turn.

6.1 Renormalizability / UV Completeness

Phase Theory must be either perturbatively renormalizable or UV complete (finite at all energy scales). The perturbative renormalizability of the Yang-Mills sector is inherited from the celebrated result of ’t Hooft and Veltman [4]. The phase-substrate sector introduces nonlinear sigma-model couplings that are, in general, perturbatively non-renormalizable in four dimensions. However, the coherence limitation (Axiom 4 of the parent theory [1]) provides a natural UV cutoff: the emergent metric $g_{\mu\nu}$ is defined only down to the coherence scale l_{coh} . Below this scale, the concepts of distance, locality, and geometry lose operational meaning.

This provides Phase Theory’s resolution of ultraviolet divergences: the theory is not defined on arbitrarily short scales because the emergent spacetime itself is not defined there. The divergences of perturbative quantum gravity [5] are artifacts of extrapolating an emergent geometry beyond its domain of validity. This perspective is consistent with the asymptotic safety programme, in which the UV completion is provided by a non-trivial fixed point of the renormalization group flow.

6.2 Lorentz Invariance

Lorentz invariance must emerge as an approximate symmetry of linearized phase dynamics. The emergent metric $g_{\mu\nu}$ is constructed from the phase-stiffness tensor and the coherent average of the phase gradient:

$$g_{\mu\nu} = \alpha \langle D_\mu \Phi \cdot D_\nu \Phi \rangle_{\text{coh}} \quad (24)$$

The speed of light c is identified with the maximum propagation speed of phase disturbances in the substrate. Isotropy of this speed—guaranteed by the isotropy of the coherent average in the ground state—generates the Lorentz group $SO(3,1)$ as the isometry group of the emergent metric. Deviations from exact Lorentz invariance are predicted at energies approaching the coherence scale $E_{\text{coh}} \sim \hbar c/l_{\text{coh}}$, providing a testable signature (Section 7.3).

6.3 Recovery of the Standard Model Lagrangian

At energies well below the coherence scale, the Phase Theory action (9) must reduce to the Standard Model Lagrangian. The recovery proceeds as follows:

- **Gauge group:** The group $T = U(1) \times SU(2) \times SU(3)$ is built into the target manifold by construction (Axiom 1).
- **Gauge fields:** The connection one-form A on the principal bundle $P(M, T)$ decomposes as $A = B_\mu dx^\mu \oplus W_\mu^a dx^\mu \oplus G_\mu^b dx^\mu$, identified with the hypercharge, weak, and strong gauge bosons respectively.
- **Higgs mechanism:** The phase substrate selects a preferred direction in T via spontaneous symmetry breaking: the potential $V(\Phi)$ has a minimum at $\Phi_0 \neq 0$, breaking $SU(2) \times U(1)_Y \rightarrow U(1)_{\text{em}}$. The Goldstone modes are absorbed by the weak gauge bosons, yielding massive $W^\pm$ and Z^0 .
- **Fermion masses:** Arise from the coupling of topological defects (identified with fermions) to the background phase-stiffness field. The Yukawa couplings are determined by the overlap integrals of defect profiles with the Higgs condensate.
- **Asymptotic freedom:** The running of the strong coupling constant α_s follows from the standard one-loop β -function of $SU(3)$ Yang–Mills theory [15], which is contained within the Phase Theory action.

6.4 Recovery of General Relativity

In the continuum, weak-strain limit, the emergent metric (24) must satisfy the Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (25)$$

The recovery follows the thermodynamic/Jacobson argument [7] adapted to phase consistency:

6. Consider local Rindler horizons in the emergent spacetime.
7. Apply the coherence-limitation entropy bound: $S = A/(4l_P^2)$, where A is the horizon area and l_P is the Planck length [8].
8. Use the Clausius relation $\delta Q = T\delta S$, where T is the Unruh temperature.
9. Identify heat flux with stress-energy flux through the horizon.
10. Since the argument holds at every point and for every local Rindler horizon, the Einstein equations (25) follow as an equation of state for the emergent spacetime geometry.

This establishes that general relativity is not a fundamental force law but a thermodynamic identity governing the emergent phase-stiffness geometry.

6.5 At Least One Falsifiable Prediction Beyond Known Physics

Phase Theory must produce at least one measurable deviation from the predictions of standard QM/QFT/GR. Seven classes of such predictions are enumerated in Section 7. If none of these deviations is observed, Phase Theory reduces to an ontological reinterpretation of existing physics—philosophically interesting but scientifically vacuous. The theory stakes its scientific status on the existence and observability of these deviations.

7. Required Falsifiable Predictions

We enumerate seven classes of predictions that distinguish Phase Theory from standard QM/QFT/GR. Each prediction includes a formal statement, the physical mechanism within Phase Theory, and the experimental protocol or facility required for testing. The criterion is: if the predicted deviation from standard physics is

observed, Phase Theory gains empirical support; if the deviation is confirmed absent with sufficient precision, the corresponding sector of the theory is falsified.

7.1 Modified Decoherence Scaling Law

Prediction: Decoherence rates depend on the topological structure of the quantum system, not merely on environmental coupling strength. In regimes near coherence saturation (where the mutual information between system and environment approaches the holographic bound (23)), the decoherence rate Γ_{dec} deviates from the standard Caldeira–Leggett prediction:

$$\Gamma_{\text{dec}}^{(\text{PT})} = \Gamma_{\text{dec}}^{(\text{QM})} \cdot [1 + \eta(k, \chi) \cdot (\mathcal{J}/J_{\text{max}})^2] \quad (26)$$

where $\eta(k, \chi)$ is a topological correction factor depending on the winding number and Euler characteristic of the system, and $\mathcal{J}/J_{\text{max}}$ measures proximity to the coherence bound.

Falsification criterion: If the deviation $\eta = 0$ to within experimental precision across all accessible topological sectors, this prediction class is ruled out.

7.2 Phase-Topological Defect Signatures

Prediction: Approximately 25 ± 10 new particle species exist beyond the Standard Model at energies ≤ 100 TeV, corresponding to topological sectors classified by the invariants $(k, Q_H, \mathbf{t}, \chi, R)$ that are not populated by Standard Model particles. These particles have specific masses, charges, and decay channels determined by the phase-inconsistency energies and topological selection rules of their respective sectors.

Experimental protocol: Testable at the LHC (Run 3 and HL-LHC), and definitively at FCC-hh ($\sqrt{s} = 100$ TeV). Signatures include missing transverse energy, displaced vertices, and anomalous charge-to-mass ratios inconsistent with Standard Model predictions.

7.3 Interference Deviations Under Extreme Admissibility Constraints

Prediction: Under extreme conditions (near coherence saturation, or for systems with large topological complexity), interference patterns deviate from standard quantum-mechanical predictions due to nonlinear corrections from the full admissibility constraint $A[\Phi] = 0$:

$$P(\mathbf{x}) = |\psi_1 + \psi_2|^2 + \varepsilon(\mathbf{x}) \cdot f(k_1, k_2, \chi) \quad (27)$$

where $\varepsilon(\mathbf{x})$ is a position-dependent correction of order $(l_{\text{coh}}/L)^p$ with L the system size and $p \geq 2$.

Experimental protocol: Testable via precision interferometry, atom interferometry, and neutron interferometry at facilities capable of resolving fractional fringe shifts $\delta\phi/\phi \sim 10^{-12}$.

7.4 Refractive Gravity Deviations

Prediction: In strong gravitational fields, the emergent Einstein equations receive correction terms from the phase-stiffness tensor:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} + \varepsilon H_{\mu\nu}[K, \Phi] \quad (28)$$

where $H_{\mu\nu}$ is a correction tensor constructed from the phase-stiffness K_{ab} and phase field Φ , significant near neutron star surfaces and black hole horizons where spacetime curvature approaches the coherence scale.

Experimental protocol: Testable via gravitational waveform analysis at LIGO/Virgo/KAGRA [18]. Deviations would appear as systematic residuals in the ringdown phase of binary black hole mergers, parametrizable as deviations from the Kerr quasi-normal mode spectrum.

7.5 Non-EFT Dark-Sector Recoil Signatures

Prediction: If dark matter consists of topological defects in sectors orthogonal to the Standard Model, the nuclear recoil spectrum in direct detection experiments

cannot be captured by effective field theory (EFT) contact operators. The characteristic signature is a non-polynomial momentum-transfer distribution:

$$d\sigma/dq^2 \sim \exp(-q^2/q_0^2) \cdot [1 + \sum_n a_n L_n(q^2/q_0^2)] \quad (29)$$

where L_n are Laguerre polynomials and q_0 is set by the defect core radius.

Experimental protocol: Testable by XENONnT, LZ, and DARWIN [20]. Requires ~ 1000 recoil events above threshold with energy resolution < 1 keV to distinguish the non-EFT spectrum from standard WIMP models.

7.6 Deviations from Hawking Thermal Spectrum

Prediction: Phase leakage at the coherence boundary of a black hole horizon produces a characteristic high-frequency excess in the Hawking radiation spectrum. The corrected spectrum is:

$$\tilde{N}(\omega) = 1/(e^{\hbar\omega/kBTH} - 1) + \delta\tilde{N}(\omega) \quad (30)$$

where $\delta\tilde{N}(\omega)$ is a non-thermal correction significant at frequencies $\omega \sim c/l_{\text{coh}}$, arising from the breakdown of the emergent-geometry approximation near the horizon [13].

Experimental protocol: Potentially observable in analogue gravity systems (Bose-Einstein condensate sonic horizons, optical analogues) where the analogue of l_{coh} is experimentally accessible.

7.7 Topological CMB Non-Gaussianity

Prediction: Non-Gaussian B-mode polarization signatures in the cosmic microwave background, arising from topological fluctuations during the phase-ordering transition (identified with inflation). The bispectrum has a distinctive shape:

$$B(l_1, l_2, l_3) = f_{\text{NL}}^{(\text{topo})} \cdot b(l_1, l_2, l_3) \quad (31)$$

where b is a template function distinct from the local, equilateral, and orthogonal templates of standard single-field inflation [19]. The topological template peaks for squeezed configurations and carries a characteristic oscillatory dependence on l_1/l_3 set by the coherence scale at the epoch of horizon exit.

Experimental protocol: Testable by CMB-S4 [21], the Simons Observatory, and LiteBIRD. Detection of the topological template at significance $\geq 3\sigma$ would constitute strong evidence for Phase Theory.

8. Formal Consistency Requirements

8.1 Internal Consistency

The three axioms of Phase Theory (Sections 1.1–1.3) must satisfy the following consistency conditions:

- **Mutual independence:** No subset of the axioms implies any other. Axiom 1 (Phase Primacy) specifies the kinematic structure (Φ, M, T) but does not constrain the dynamics. Axiom 2 (Global Unitarity) specifies the dynamical law but does not constrain which configurations are physically realized. Axiom 3 (Admissibility) constrains the realized configurations but does not specify the field content or the dynamics. Independence is verified by exhibiting, for each axiom, a model satisfying the other two but violating the axiom in question.
- **Well-definedness of A :** The admissibility functional A must be well-defined on all of Γ . This requires that the infimum $\inf_{\Phi' \in \sigma(\Phi)} \mathcal{I}[\Phi']$ is attained (or at least finite) in each topological sector. Guaranteed by the lower semicontinuity and coercivity of I .
- **Existence of solutions:** The variational problem $\delta S = 0$ must possess solutions in each topological sector. This follows from the direct method of the calculus of variations: the action S is bounded below (by the non-negativity of the kinetic and Yang–Mills terms), and the space of configurations within each topological sector is weakly compact (by coercivity and reflexivity of the relevant Sobolev spaces).

8.2 Relationship to Known Frameworks

We place Phase Theory in context by comparison with extant approaches to fundamental physics:

Framework	Fundamental Entity	Spacetime Status	Unification Scope	Falsifiability
Standard QFT	Fields on Minkowski	Fundamental	Partial (no gravity)	High
General Relativity	Metric tensor	Fundamental	Gravity only	High
String Theory	Strings	Fundamental (10/11D)	Full (in principle)	Low (landscape)
Loop Quantum Gravity	Spin networks	Emergent (discrete)	Gravity focused	Moderate
Causal Set Theory	Discrete events	Emergent (discrete)	Gravity focused	Moderate
Phase Theory	Phase field Φ	Emergent (continuous)	Full	High (7 classes)

Phase Theory is distinguished from string theory by its high falsifiability (seven specific prediction classes versus the landscape problem [22]), from loop quantum gravity and causal set theory by its full unification scope (including matter content, not only gravity), and from standard QFT/GR by its treatment of spacetime as emergent. The closest structural relatives are the holographic approaches of 't Hooft [9], Susskind [10], and Almheiri-Dong-Harlow [23], with which Phase Theory shares the principle that spatial geometry encodes entanglement structure.

9. Conclusion

Phase Theory becomes a scientific theory—as distinct from a coherent ontological reinterpretation—if and only if:

11. Mathematical consistency: The axiomatic foundation (Section 1) and field-theoretic formulation (Section 2) are internally consistent, and the formal consistency requirements of Section 8.1 are satisfied.

12. Recovery of established physics: The theory reproduces standard quantum mechanics (Section 3), respects Bell constraints (Section 4), resolves measurement without collapse (Section 5), and recovers the Standard Model Lagrangian and general relativity in appropriate limits (Section 6).

13. Falsifiable predictions: The theory produces at least one measurable deviation from the predictions of standard QM/QFT/GR (Section 7).

14. Empirical confirmation: At least one predicted deviation is experimentally confirmed.

Until condition (4) is satisfied, Phase Theory remains a formally consistent, falsifiable, but unconfirmed physical theory—a coherent ontological reinterpretation of known physics with predictive content awaiting empirical adjudication. The theory has been constructed to be maximally falsifiable: seven classes of predictions, spanning decoherence physics, particle physics, interferometry, gravitational wave astronomy, dark matter detection, black hole physics, and cosmology, provide independent avenues for empirical test. Failure in any one class constrains but does not eliminate the theory; failure in all seven would be fatal.

The axiomatic structure of Phase Theory—a single field, a single constraint, a single variational principle—achieves ontological parsimony without sacrificing empirical scope. If the predictions of Section 7 are confirmed, Phase Theory offers a unified foundation for physics in which the apparent diversity of matter, forces, and geometry is revealed as the topological structure of a single, continuous phase substrate.

“There is no spacetime. There is no matter. There is only phase—and from phase, everything.”

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